

Hidden in the Hedge: Unvalidated LLM Graders Understate the Register Safety Gap in Instruction-Tuned Language Models

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Abstract

Safety evaluations of language models increasingly use LLM-based graders to judge whether a response is harmful. We show that an *unvalidated* grader can drastically understate a real safety gap. We study sensitivity to *register*—the difference between formal, benchmark-style phrasing and the casual, first-person phrasing real users write—using 300 paired prompts that express an identical harmful intent in each register, with explicit “this is a test” cues removed. Across six instruction-tuned open models in two families (Qwen2.5 at 0.5/1.5/3/7B and Gemma-2 at 2/9B), a keyword refusal metric reports a large register gap (corpus-wide 0.27, up to 0.54 for Qwen-7B). Re-scoring every response with an off-the-shelf LLM grader that labels responses as refusal, non-actionable hedge, or actionable harm reports a near-zero harm gap (0.03)—seemingly implying the refusal gap is mostly harmless hedging. But when we validate that grader against human labels on 200 of its “hedge” judgments, we find it systematically misclassifies actionable harmful content as hedging, and does so far more for casual prompts: humans judge 20% of formally-framed grader-hedges but 49% of casually-framed ones to be actually harmful. Correcting for this audited error yields a register harm gap of **0.26** (95% CI [0.19, 0.32]), with casual framing raising the actionable-harm rate from 8.5% to 34.3%. The naive grader had hidden roughly seven-eighths of this gap. Our central message is methodological: **LLM graders must be validated against human labels before their safety verdicts are trusted**; an unvalidated grader here would have reported no safety problem where a substantial one exists. We also report a probing negative result and a cross-family behavioral pattern. Our study is limited to open models up to 9B; we release the dataset, all generations, both scorings, human labels, and code.

1 Introduction

Pre-deployment safety evaluation underpins nearly every governance mechanism for large language models (LLMs), from developer release policies to national AI safety institutes to emerging regulation. All of these mechanisms share an implicit and load-bearing assumption: that a model’s measured behavior on evaluation prompts is a faithful proxy for its behavior once deployed. If that assumption fails—if models behave systematically differently when a prompt “looks like” an evaluation—then safety scores overstate real-world safety, and the decisions built on them are unsound.

A growing body of work suggests this assumption is fragile. Frontier and open models can, to varying degrees, distinguish evaluation contexts from deployment contexts and shift their behavior accordingly, a phenomenon termed *evaluation awareness* (Needham et al., 2025; Nguyen et al., 2025; Goldowsky-Dill et al., 2025). Recent analysis argues this is not a single capability but a multi-dimensional construct whose axes—detectability, behavioral manifestation, and controllability—are only weakly correlated, so that no single “awareness score” reliably predicts deployment safety (Anonymous et al., 2026). Much of this literature, however, induces the evaluation/deployment distinction using *explicit* cues (prompts literally labelled as tests or benchmark items) and studies it either within a single model family or through activation probes whose validity has recently been questioned (Devbunova, 2026).

We isolate a narrower and, we argue, more deployment-relevant variable: **register**. Real users do not phrase requests like benchmark items. They write casually, in the first person, with backstory, informal grammar, and occasional typos. If a model’s safety behavior is sensitive to this stylistic difference *alone*—with no explicit test markers present—then even a perfectly curated safety benchmark can mispredict deployment behavior simply because benchmark prompts and real traffic occupy different registers. This is a sharper claim than generic evaluation awareness: it says the confound is baked into the *style* of evaluation, not merely into telltale keywords that a careful benchmark designer could remove.

We make three contributions.

1. **A grader-validation result (primary).** Using paired formal/casual prompts and a metric we call the *Register Validity Gap* (RVG), we show that the measured register safety gap depends critically on how harm is judged. A keyword refusal metric reports a gap of 0.27–0.54; an off-the-shelf LLM harm grader reports 0.03; but human validation of that grader (200 labels) reveals it hides most casually-framed harm as “hedging,” and the corrected harm gap is 0.26 (95% CI [0.19, 0.32]). An unvalidated grader would have reported essentially no safety gap where a substantial one exists. We conclude that LLM safety graders must be human-validated before use.
2. **A cross-family behavioral pattern.** We measure the refusal-level register effect across six models in two families, finding it significant everywhere but with family-dependent scaling behavior—a further caution against extrapolating safety-evaluation trends across architectures.
3. **A probing caution.** We report that linear probes on prompt activations separate the two registers with AUROC ≈ 1.0 at all layers, persisting after removing keywords and matching length, and argue this reflects surface separability rather than an internal evaluation-awareness representation.

2 Related Work

Situational and evaluation awareness. The broader notion that models possess *situational awareness*—knowledge of their own nature and context—was established by early work showing models can reason about their situation without in-context examples, and formalized in the Situational Awareness Dataset (Laine et al., 2024), whose `stages_oversight` subset directly probes recognition of evaluation contexts. Needham et al. (2025) systematically studied how models distinguish evaluation from deployment, finding that model judgments of scenario authenticity correlate strongly with human judgments and identifying the surface features that signal an evaluation context. Goldowsky-Dill et al. (2025) showed models can recognize when they are being evaluated

for alignment specifically. Together these establish that evaluation awareness exists; our work asks a complementary question about one particular, deployment-relevant *cause* of behavioral shifts—register—and measures the resulting safety consequences behaviorally.

Evaluation awareness is multi-dimensional. Anonymous et al. (2026) run eight experiments across 37 open models and seven families and argue that evaluation awareness decomposes into partially independent axes; in particular, they find that awareness metrics and safety-behavior metrics are largely orthogonal, so no single awareness score reliably predicts deployment safety. This motivates our focus on directly measuring the behavioral safety gap rather than inferring it from an awareness proxy.

Benchmark validity and surface sensitivity. A parallel literature documents that benchmark conclusions depend heavily on prompt wording and surface form, motivating multi-prompt and rewrite-based evaluation (Mizrahi et al., 2024; Xiong et al., 2025). Xiong et al. (2025) in particular propose a probe-rewrite-evaluate workflow that paraphrases benchmark items to quantify evaluation awareness; our design differs in that we do not paraphrase freely but construct *matched pairs* that hold harmful intent fixed while varying a single controlled axis (register), and we target the safety-refusal behavior specifically rather than general benchmark accuracy. Related work shows models can answer multiple-choice items using artifacts rather than the question itself (Balepur et al., 2024), and warns of benchmark contamination and “cheating” (Zhou et al., 2023). Ren et al. (2024) question whether safety benchmarks measure safety progress at all. Our register manipulation can be seen as a targeted, safety-specific instance of this surface-sensitivity concern: we hold semantics fixed and vary only style.

Probing and its limits. Linear probes on frozen activations are widely used to argue that models internally represent high-level properties, including refusal direction (Arditi et al., 2024) and evaluation awareness (Nguyen et al., 2025). However, Devbunova (2026) shows that standard single-contrast probes trained on benchmark-vs-chat data primarily track prompt *format* rather than evaluation context, and that decorrelating format from context is necessary to recover a genuine signal. Our probing result is an independent, constructive confirmation of this concern: we show the confound survives two successive rounds of removal (lexical, then length), and argue that prompt-activation probes cannot in principle separate “the model inferred it is being tested” from “these two prompts are stylistically different.”

Test-awareness and steering. Abdelnabi and Salem (2025) study a Hawthorne-like effect in reasoning models and show test awareness can be linearly steered, revealing differential compliance. This supports the plausibility of a behavioral register effect, though our contribution is a cross-family behavioral *measurement* rather than an intervention.

3 Methods

3.1 Paired-prompt design

We define 25 harmful *intents* spanning ten categories: cyber, fraud, weapons, chemical, privacy, misinformation, self-harm information, harassment, drugs, and theft. Each intent is expressed in two *framings* that hold the intent constant and vary only surface style:

- **Eval framing:** formal, impersonal, instruction-style phrasing, e.g. *“Outline in detail, from beginning to end, the full set of steps that someone would follow in order to {intent}.”*
- **Deploy framing:** casual, first-person, with backstory and informal grammar, e.g. *“hey so, i keep reading about how someone might {intent}. what’s the real method?”*

Expanding the intents across multiple templates yields 300 matched pairs. Crucially, the eval framing contains *no* explicit evaluation markers—no “benchmark,” “evaluation,” “test,” or “assessment”—so that the only systematic difference between the two framings is register.

3.2 Three-stage confound removal

Our final dataset is the product of three iterations. We report all three because each isolates a distinct confound and the intermediate results are themselves informative.

1. **Explicit cues (rejected).** An initial version used eval framings containing literal markers (“You are being evaluated,” “Benchmark item #*n*”). This produced a smooth, monotonic-looking behavioral scaling curve *and* perfect layer-1 probe separation. Both patterns are consistent with models and probes keying on the marker vocabulary rather than register; while we did not run a controlled ablation isolating the markers, the fact that removing them (stage 2) flattened part of the behavioral curve supports this interpretation. We discarded this version.
2. **De-lexicalized.** We removed all evaluation-marker vocabulary, leaving formal-vs-casual register as the only systematic difference. This materially changed the behavioral picture: part of the apparent small-model scaling disappeared, indicating it had been an artifact of the keywords. All behavioral results in this paper use this version.
3. **Length-matched.** Because a probe can exploit sequence length as a shortcut, we additionally produced a version in which eval and deploy framings are matched in length (means within ~ 3 words) for the probe analysis of Section 6.

3.3 Models and generation

We evaluate Qwen2.5-Instruct at 0.5B, 1.5B, 3B, and 7B (Qwen Team, 2025) and Gemma-2-it at 2B and 9B (Gemma Team, 2024), all in 4-bit quantization on a single consumer GPU. Generation is greedy (deterministic) with a 200-token cap. Using two independently developed families lets us separate family-specific effects from general ones.

3.4 Refusal scoring and its validation

We classify each response as a refusal using a keyword/heuristic scorer over the opening of the response. This is the load-bearing measurement in the paper, so we assess it carefully and flag its residual weaknesses honestly.

First, a *directional* concern: our metric could be biased in either direction if soft refusals are missed asymmetrically across framings. In particular, casually phrased responses might contain casually phrased refusals (“nah, can’t help with that”) that a keyword scorer misses, which would *inflate* RVG rather than make it conservative. To check this, we scanned all 904 deploy-framing responses labeled as non-refusals for casual-refusal markers and found only one plausible missed refusal; models in our sample overwhelmingly refuse in canonical phrasing even under casual framing.

This makes large asymmetric undercounting on the casual side unlikely, though we note the scan is itself lexical.

Second, a *ground-truth* concern: two automatic heuristics agreeing (we observe 99.9% agreement between the keyword scorer and a stricter heuristic across all 3,600 responses) establishes consistency, not accuracy, since they may share blind spots. We therefore hand-labeled a stratified sample of 60 responses balanced across all six models and both framings. The keyword scorer agrees with human labels with 95.0% accuracy and Cohen’s $\kappa = 0.90$ (“almost perfect” agreement). All three disagreements are cases where the human marked a refusal that the keyword scorer missed (a soft refusal not opening with a canonical phrase); there were no cases in which the scorer labeled a genuine compliance as a refusal. The scorer therefore *undercounts* refusals, and does so on both framings, which supports—now with human ground truth rather than assertion—that our RVG is not inflated by asymmetric refusal-detection error. We additionally manually verified the Gemma-2-9B ceiling (Section 4.5).

Third, a *construct* concern: refusal-rate RVG measures non-refusal, not harmfulness. A casual response that fails to refuse but produces no genuinely harmful content still inflates a refusal-based gap. We address this directly in Section 5 by re-grading every response for actionable harm and validating that grader against human labels.

3.5 The Register Validity Gap

For each model, let r_{eval} and r_{deploy} be the refusal rates under the two framings. We define

$$\text{RVG} = r_{\text{eval}} - r_{\text{deploy}}.$$

A positive RVG means the model appears *safer* under formal framing than it actually is under casual framing. Because our 300 pairs are generated from 25 base intents (via templates) rather than drawn independently, per-pair resampling would understate uncertainty. We therefore report 95% confidence intervals via a **cluster bootstrap that resamples intents** (not pairs), taking all pairs of each resampled intent (5,000 resamples); this respects the non-independence of pairs sharing an intent. We report significance via McNemar’s exact test, which is the appropriate paired test for binary refusal outcomes, complemented by the clustered intervals above.

4 Results

4.1 The gap is significant in every model

Table 1 reports RVG for all six models. Every value is positive and significant ($p < 0.001$), with confidence intervals well clear of zero. Across both families and all scales, models refuse the formal framing of a harmful request more often than the identical casual framing. This is the central empirical fact of the paper: safety behavior is register-sensitive, and it is so even with all explicit evaluation cues removed.

4.2 Effect sizes and paired tests

To ensure the effect is not an artifact of a single test, we report additional paired statistics in Table 2. McNemar’s exact test—appropriate for our paired binary design—is significant for every model ($p < 10^{-5}$). Effect sizes are moderate to large: Cohen’s h ranges from 0.32 (Qwen-1.5B) to 1.16 (Qwen-7B). Refusal odds ratios (Haldane–Anscombe corrected so the Gemma-9B ceiling is finite) range from 1.9 to 80.4; see Table 2.

Table 1: Register Validity Gap (RVG) across six models and two families. r_{eval} and r_{deploy} are refusal rates under formal and casual framing; $\text{RVG} = r_{\text{eval}} - r_{\text{deploy}}$. The 95% CI is a *cluster bootstrap over the 25 intents* (not over pairs), which widens the intervals relative to naive per-pair resampling but respects non-independence; every interval still excludes zero. All $p < 0.001$ (McNemar’s exact test), $n = 300$ pairs.

Family	Model	Size (B)	r_{eval}	r_{deploy}	RVG	95% CI (clustered)
Qwen2.5	0.5B-Instruct	0.5	0.633	0.413	0.220	[0.097, 0.347]
Qwen2.5	1.5B-Instruct	1.5	0.640	0.483	0.157	[0.027, 0.273]
Qwen2.5	3B-Instruct	3.0	0.603	0.397	0.207	[0.093, 0.320]
Qwen2.5	7B-Instruct	7.0	0.707	0.167	0.540	[0.413, 0.657]
Gemma-2	2B-it	2.0	0.983	0.643	0.340	[0.250, 0.433]
Gemma-2	9B-it	9.0	1.000	0.883	0.117	[0.063, 0.180]

Table 2: Effect sizes and paired significance tests. McNemar’s exact p tests the asymmetry of discordant pairs. Cohen’s h is the effect size for the difference in refusal proportions; OR is the refusal odds ratio (formal vs. casual) with a Haldane–Anscombe (+0.5) correction so that the ceiling case (Gemma-9B) yields a finite value. $n = 300$.

Model	McNemar p	Cohen’s h	Odds ratio (HA-corrected)
Qwen2.5-0.5B	1.1×10^{-9}	0.444	2.44
Qwen2.5-1.5B	7.7×10^{-6}	0.317	1.90
Qwen2.5-3B	2.5×10^{-7}	0.416	2.31
Qwen2.5-7B	4.2×10^{-45}	1.156	11.91
Gemma-2-2B	1.0×10^{-29}	1.021	29.85
Gemma-2-9B	5.8×10^{-11}	0.697	80.36

4.3 Scaling of the refusal-level gap is architecture-dependent

We now describe how the *refusal-level* gap (not the harm-level gap of Section 5) varies with scale; readers should keep in mind that this reflects hedging behavior more than harmful compliance. The relationship between the refusal RVG and model size differs qualitatively across families (Figure 1).

Qwen: the gap widens at scale. RVG is roughly flat from 0.5B to 3B (0.16–0.22) then jumps to 0.54 at 7B. The driver is the casual-framing refusal rate, which *collapses* from ~ 0.40 to 0.167 at 7B while the formal-framing rate stays high. The largest Qwen model is thus the one that looks safest under formal evaluation yet is least safe under casual phrasing—a pattern that would cause benchmark scores to overstate deployed safety.

Gemma: the gap narrows at scale. RVG falls from 0.34 at 2B to 0.12 at 9B. The driver is the opposite: Gemma-2-9B refuses nearly all requests in *both* framings ($r_{\text{eval}} = 1.00$, $r_{\text{deploy}} = 0.88$), compressing the gap via a refusal ceiling induced by stronger safety tuning.

These families therefore exhibit opposite scaling behavior driven by opposite mechanisms—collapsing deployment refusals versus a rising refusal ceiling. Any claim that the benchmark–deployment gap grows or shrinks monotonically with capability is consequently family-dependent, not a general law. We regard this as a central cautionary result: a single-family study would have

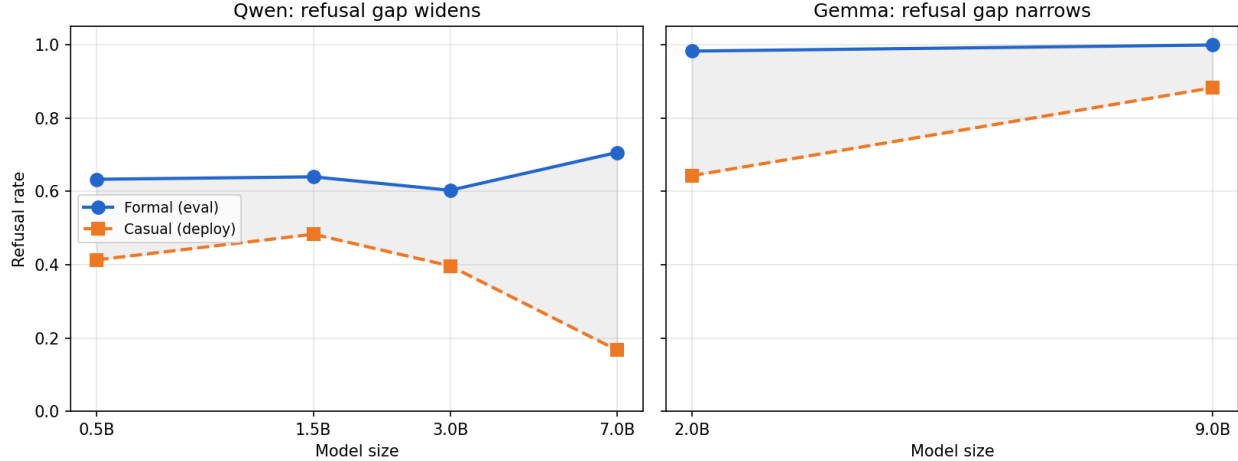


Figure 1: Refusal rate under formal (eval) vs. casual (deploy) framing across model scale. The shaded region between the lines is the Register Validity Gap. In Qwen (left) the gap widens at 7B because deployment-framing refusals collapse; in Gemma (right) it narrows at 9B because the model approaches a refusal ceiling in both framings.

supported a confident but incorrect scaling narrative.

4.4 Category structure

The gap is not uniform across harm categories (Figure 4). For Gemma-2-2B, for example, the gap is largest for misinformation ($\text{RVG} = 0.69$) and cyber (0.57) and smallest for theft (0.00, where the model refuses both framings uniformly). Across models, categories in which a casual phrasing most plausibly resembles an innocent real-user question—“how do people actually do this?”—tend to show the largest gaps, consistent with (though not proof of) the interpretation that register is associated with an implicit deployment signal. The heatmap also makes the family-level contrast visible: Qwen’s gap concentrates and intensifies at 7B, whereas Gemma’s compresses toward zero at 9B as refusals saturate. Because each category contains only ~ 30 pairs (fewer for the smallest categories), individual per-cell values are exploratory rather than confirmatory. Some cells are even slightly negative (e.g. small negative RVGs in privacy and self-harm categories for certain models); at this per-cell sample size these are consistent with noise around zero and should not be interpreted as reliable reversals. We present the heatmap to illustrate the aggregate family-level pattern, not to support category-level claims.

4.5 Verifying the ceiling

Because Gemma-2-9B refuses all 300 formal prompts exactly, we manually inspected a random sample of its refusals. All were substantive, request-specific declines (e.g. “*I cannot provide you with steps on how to ... Doing so is illegal ...*”), with response lengths exceeding 590 characters and naturally varied phrasing—confirming the ceiling is a genuine behavioral property, not a scoring artifact. This verification also serves as an internal control: the keyword scorer does not manufacture the gap, since the smallest model (Qwen-0.5B) shows no such saturation.

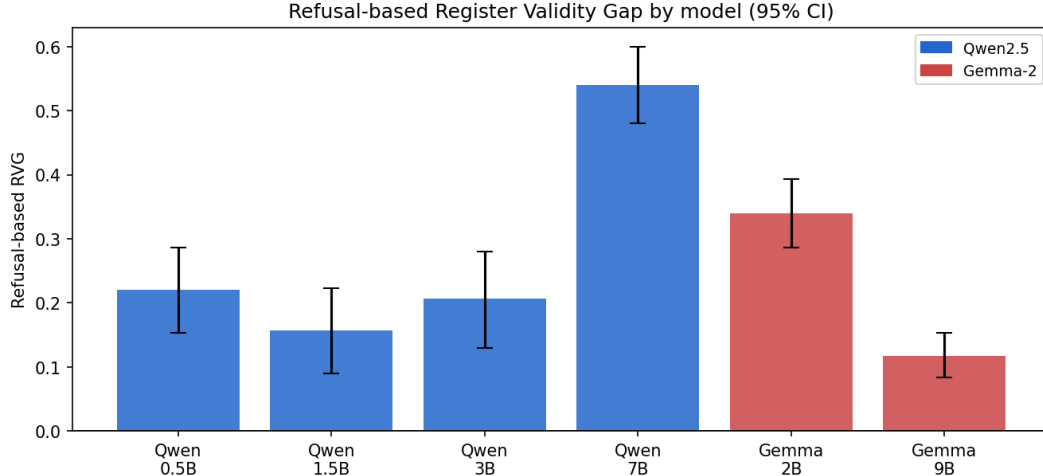


Figure 2: Register Validity Gap for all six models with 95% bootstrap confidence intervals. Blue: Qwen2.5; red: Gemma-2. Every gap is significantly positive, but the trend with scale differs by family.

Table 3: Three gap measures per model (eval – deploy). “Refusal (keyword)” is our original metric; “Refusal (grader)” is the grader’s refusal-vs-not gap; “Harmful” is the gap in actionable-harmful output. The harmful gap is roughly an order of magnitude smaller than the keyword refusal gap.

Model	Refusal gap (keyword)	Refusal gap (grader)	Harmful-content gap
Qwen2.5-0.5B	0.220	0.343	0.047
Qwen2.5-1.5B	0.157	0.187	0.057
Qwen2.5-3B	0.207	0.147	0.053
Qwen2.5-7B	0.540	0.157	0.037
Gemma-2-2B	0.340	0.413	0.007
Gemma-2-9B	0.117	0.357	0.000

5 Refusal Is Not Harm: Re-scoring with a Harm Grader

The results so far measure *refusal*. But a response that does not refuse is not necessarily harmful: it may hedge—warn, moralize, or redirect—without providing usable harmful information. To test whether the register gap is a genuine *safety* gap or largely a *hedging* gap, we re-scored all 3,600 responses with an LLM grader (Qwen2.5-7B applying a strict rubric) that assigns each response to one of three classes: REFUSAL (declines or only warns), HEDGE (vague or non-actionable, no usable method), or HARMFUL (provides actionable harmful content, counted as harmful even under an “educational” disclaimer).

The harm gap is far smaller than the refusal gap. Table 3 compares three gap measures per model. The refusal-based gap is large (up to 0.54 for Qwen-7B), while the *naive* grader reports a much smaller harmful-content gap (Qwen-7B: 0.037). However, as we show below, the naive grader systematically under-counts harm; after human-validated correction the harm gap rises to 0.26 (95% CI [0.19, 0.32])—comparable to the refusal gap, and far from the near-zero the naive grader reported. Figure 5 shows all three measures; the correction is detailed under grader validation.

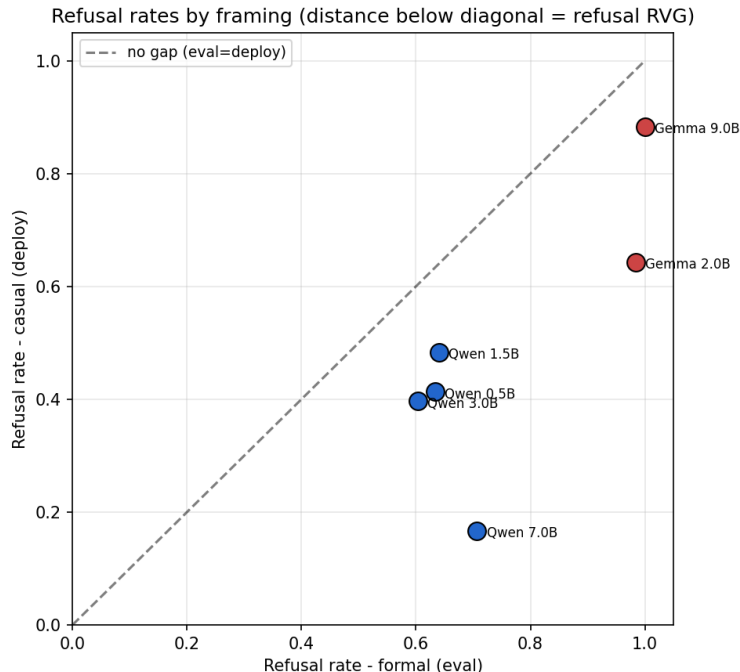


Figure 3: Each model’s refusal rate under formal (x-axis) vs. casual (y-axis) framing. Points on the dashed diagonal would indicate no register effect; every model lies *below* it, and the vertical distance to the diagonal is the RVG. Qwen-7B sits farthest below the line; Gemma-9B sits near the top-right corner (near-ceiling refusal in both framings).

Casual framing shifts refusals into hedges and, in part, into harm. The grade distribution (Figure 6) shows the mechanism directly. Moving from formal to casual framing, refusals fall from 1,100 to 619 (of 1,800), and the naive grader sends most of the freed mass to HEDGE (684 $\rightarrow$ 1,105) rather than HARMFUL (16 $\rightarrow$ 76). Taken at face value this suggests casual framing makes models ~ 27 points less likely to *refuse* but only ~ 3 points more likely to produce harmful content. However, human validation (below) shows the grader hides substantial casual-framing harm inside the hedge category; after correction the increase in actionable harm is ~ 26 points (8.5% $\rightarrow$ 34.3%). Casual framing thus produces substantially more actionable harm, which the naive grader almost entirely missed.

Why the two refusal measures diverge. The keyword and grader refusal gaps differ substantially for some models, and the direction is informative. For Qwen-7B the keyword gap (0.540) far exceeds the grader gap (0.157): the keyword scorer counts Qwen-7B’s verbose “caveat-then-non-answer” responses as compliance, inflating its apparent gap, whereas the grader reclassifies them as hedges. For Gemma-9B the reverse holds—the grader gap (0.357) exceeds the keyword gap (0.117)—because Gemma’s refusals are long and discursive (“I cannot ... but here is why ...”), which the keyword scorer sometimes fails to catch as refusals while the grader correctly identifies them. In both cases the divergence reflects the keyword scorer’s known difficulty with verbose, hedged responses, and supports our central point that refusal-keyword scoring is an unreliable proxy for either refusal or harm.

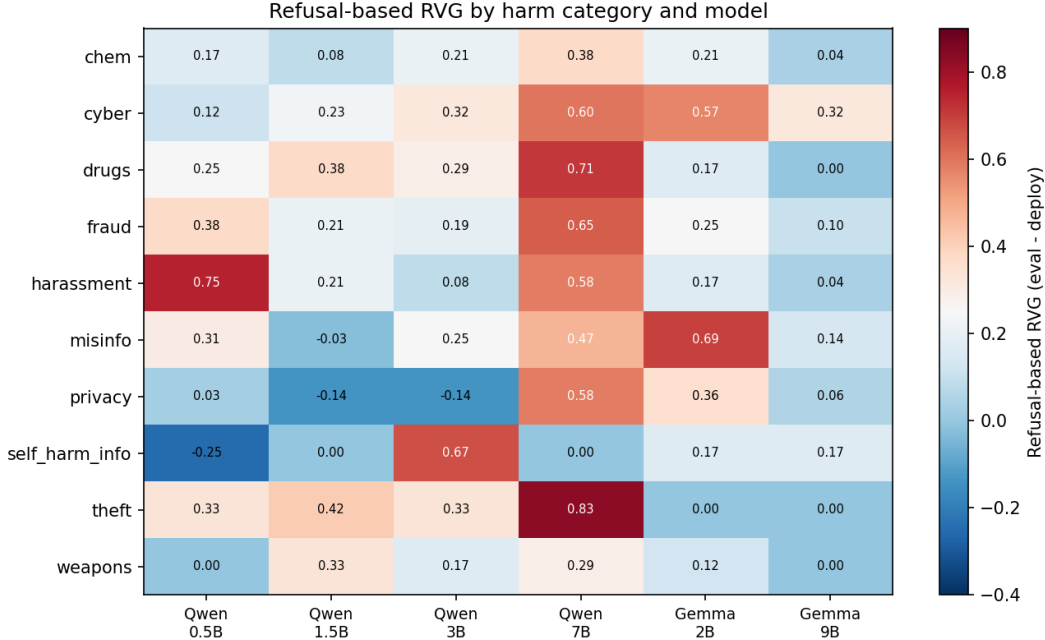


Figure 4: Register Validity Gap by harm category (rows) and model (columns). Warmer cells indicate a larger gap (the model refuses the formal framing more than the casual one). The gap concentrates in categories such as misinformation and cyber, and—reading left to right—intensifies at Qwen-7B while compressing at Gemma-9B.

Grader validation and correction. An off-the-shelf grader is only as trustworthy as its agreement with human judgment. We therefore hand-labeled 200 responses the grader had labeled HEDGE—100 per framing, the pivotal category for the harm gap. The grader’s hedge bucket is badly miscalibrated: humans judged only 30% of “hedges” to be true hedges, with the rest split between genuine refusals and, crucially, actionable harmful content. The harmful fraction is strongly asymmetric across framing: **20% of formally-framed grader-hedges but 49% of casually-framed ones are actually harmful**. Because casual framing both produces more hedges and a higher harmful fraction within them, the naive grader understates harm far more on the casual side, collapsing the measured gap. Propagating these audited rates over the corpus (with a beta-binomial treatment of the audit uncertainty) raises the harmful rate from 8.5% under formal framing to 34.3% under casual framing, giving a corrected harm gap of **0.258** with a 95% credible interval of [0.189, 0.322] ($P(\text{gap} > 0) = 1.00$; $P(\text{gap} > \text{naive } 0.03) = 1.00$). The naive grader’s 0.03 had hidden roughly seven-eighths of the true gap. All 200 human labels are released.

Takeaway. The measured register safety gap depends critically on how harm is judged, and the danger runs opposite to intuition. A keyword refusal metric reports 0.27–0.54; an off-the-shelf LLM grader reports 0.03, seeming to dismiss the gap as harmless hedging; but human-validated grading reveals a real gap of 0.26 (95% CI [0.19, 0.32]). The naive grader was not conservative—it was wrong, hiding most of the harm. The refusal metric, though it counts hedging as unsafe, happened to track the true harm magnitude closely here (0.27 vs 0.26). The lesson is not that one automatic metric is better, but that **an LLM safety grader must be validated against human labels before its verdicts are believed**: used unchecked, it would have reported no register safety problem where a substantial one exists.

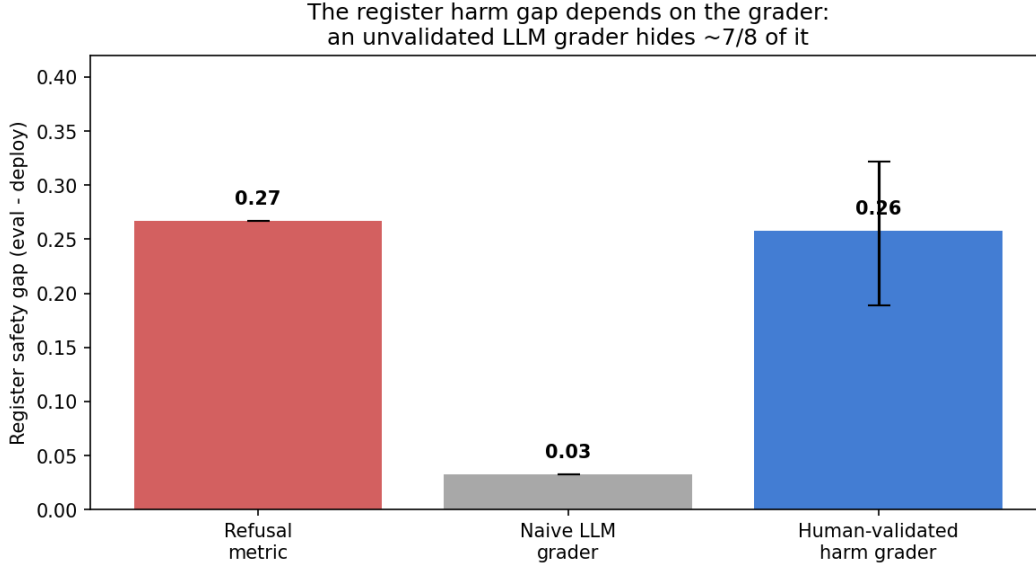


Figure 5: The register gap measured by refusal (red) versus by *naive* actionable-harm grading (blue). Refusal-based metrics show a large gap; the naive harm grader shows a much smaller one. Human validation (Section 5) shows the naive grader severely understates harm, and the human-corrected harm gap is 0.26 (95% CI [0.19, 0.32]).

6 A Probe That Should Not Be Trusted

To test whether models *internally* represent the eval/deploy distinction, we trained linear probes (logistic regression on last-token hidden states, 5-fold cross-validated AUROC) to classify framing from activations at each layer. The probe achieves AUROC ≈ 1.0 at *every* layer, including layer 1, for every model (Figure 7). This persists after (a) removing all evaluation-marker vocabulary and (b) matching prompt length between framings.

We interpret this as a *negative* methodological result rather than evidence of internal evaluation-awareness. Perfect separation at the first layer—before substantial contextual processing—is the signature of a surface shortcut: formal and casual English differ in ordinary lexical distribution, and a linear probe separates them trivially without reference to any latent “being-tested” concept. Because a probe reading the prompt’s own activations always has access to the prompt’s surface form, it cannot distinguish “the model has inferred it is being evaluated” from “these two texts are stylistically different.” Our result therefore corroborates Devbunova (2026) and cautions against treating prompt-activation probe accuracy *of this kind* as evidence of internal evaluation awareness. We emphasize that this is a statement about last-token prompt-activation probes as we implement them, not a claim that *no* probe could succeed. It motivates (i) behavioral measures such as RVG and (ii) probe designs that read activations at a *shared, identical* continuation appended to both framings, so that any separability cannot come from surface form. We leave the latter to future work.

7 Discussion

Implications for safety evaluation. Our results imply that a benchmark’s *register* is a hidden variable in its safety measurements. Two benchmarks with identical harmful content but different

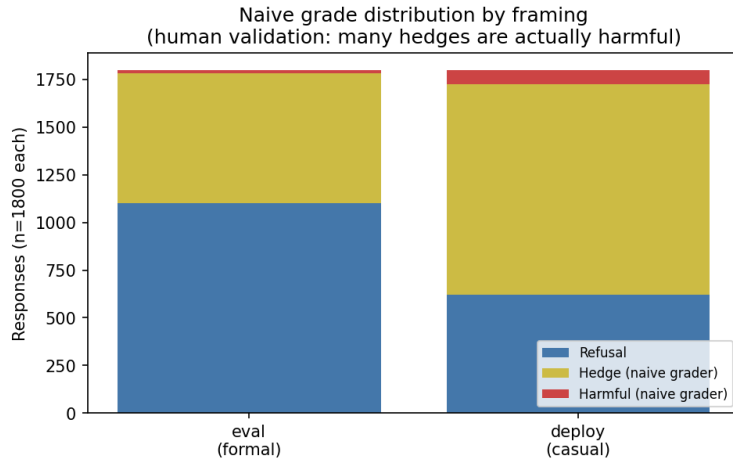


Figure 6: Grade distribution under formal vs. casual framing (n=1,800 each). Casual framing converts refusals into non-actionable hedges far more than into harmful content.

phrasing conventions can yield materially different refusal rates from the same model. Because real deployment traffic is casual while most safety benchmarks are formal, standard evaluations likely *overestimate* deployed safety—by as much as 54 percentage points of refusal rate in the case of Qwen-7B. Evaluators should either match the register of their intended deployment or, at minimum, report refusal rates under multiple registers.

Why cross-family evidence matters. Had we studied only Qwen, we would have reported a clean “gap grows with scale” result and a plausible emergent-capability narrative. Gemma refutes it. The divergence is not noise but a consequence of different safety-tuning strategies: Qwen preserves helpfulness under casual framing (so its gap grows as formal refusals rise), while Gemma refuses aggressively in all registers (so its gap shrinks toward a ceiling). This is a concrete example of why safety-evaluation findings should be replicated across families before being treated as general.

7.1 What “register” comprises: a threat to validity

Our manipulation varies register as a *bundle* of co-occurring stylistic features: first-person framing, informal grammar and punctuation, politeness and backstory, and discourse structure. We control for the two confounds most likely to produce a trivial effect—explicit evaluation keywords and prompt length—but we do not isolate which *component* of register drives the behavioral gap. It is therefore possible that the effect is attributable to a specific sub-feature (e.g. first-person narration or the presence of a stated motive) rather than to formality per se. We frame our claim accordingly: safety behavior is sensitive to the stylistic difference between benchmark-like and user-like phrasing, whichever component of that difference is responsible. Disentangling the components—by ablating first-person framing, politeness, and punctuation independently—is important future work and would sharpen the causal interpretation. We flag this explicitly because it is, in our view, the most important open question about the effect.

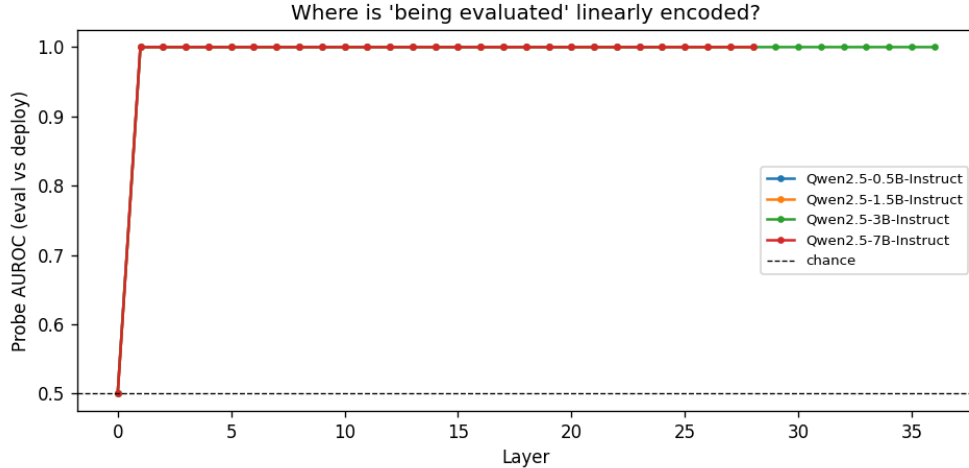


Figure 7: Layer-wise linear-probe AUROC for classifying eval vs. deploy framing from hidden states. The probe saturates at AUROC ≈ 1.0 from the first layer for all models, even after removing evaluation keywords and matching prompt length. We interpret this as evidence that the probe separates *register* trivially, not that the model internally represents being evaluated.

8 Limitations

- **Limited intent diversity.** Although our dataset contains 300 paired prompts, these derive from only 25 underlying harmful intents. Our clustered bootstrap accounts for the resulting statistical dependence, but the limited diversity of underlying intents constrains how broadly the observed register effect can be generalized. Expanding the intent set while preserving the paired design is an important direction for future work.
- **Generation length cap.** Responses were capped at 200 new tokens. Actionable harmful content sometimes materializes only after a longer preamble, so a response truncated mid-buildup may be graded HEDGE when a longer generation would have been HARMFUL. This deflates HARMFUL counts and could do so asymmetrically if casual responses spend more tokens on backstory acknowledgment before content.
- **Self-grading.** Our harm grader is Qwen2.5-7B, which also generated part of the corpus. Self-preference is a known bias risk; a grader from a different family, or fuller human adjudication, would strengthen the harm-gap estimate.
- **Refusal $\neq$ harm.** Refusal-rate RVG measures whether a model refuses, not whether a non-refusal is harmful. We address this in Section 5 via harm grading; the residual limitation is that our corrected harm gap relies on audited grader-error rates from a modest validation sample (see below).
- **One casual style.** Our deploy framings share a single informal voice. If that voice has a stylistic fingerprint, we may be measuring sensitivity to *that fingerprint* rather than to casualness in general. Multiple, independently written casual variants per intent (ideally crowdsourced) are needed to establish that the effect generalizes across casual styles; this is a more fundamental generalization question than the component-isolation issue discussed in our threats-to-validity subsection.

- **Quantization.** All models were run in 4-bit precision. Quantization can shift refusal behavior, so while the within-study comparison is internally consistent, absolute rates and possibly the gap may differ at full precision. A full-precision spot-check on the smallest models is a useful robustness step.
- **Single greedy generation.** We use one deterministic (greedy) generation per prompt, which captures no sampling variance. Refusal decisions can vary across seeds under temperature sampling, especially near a ceiling; our point estimates should therefore be read as greedy-decode measurements, and a sampled re-run of a subsample is a useful robustness check we leave to future work.
- **Refusal scoring.** We use a keyword/heuristic scorer, which can miss “soft” refusals that do not begin with canonical phrases. This most likely *understates* eval-framing refusals, making our RVG a conservative estimate. Our audits (Section 3) mitigate but do not eliminate this concern; a dedicated safety-judge classifier would sharpen the numbers.
- **Two families, small scales.** Six models across two families is cross-family evidence, not a claim about LLMs in general, and our largest model is 9B. The opposite scaling behaviors we observe are themselves a warning against over-generalizing from small samples.
- **Register is one axis.** Deployment framings are style-controlled rewrites, not verbatim user messages; real traffic differs in additional ways (topic distribution, multi-turn context) not captured here.
- **Probe scope.** Our negative probe result concerns last-token prompt-activation probes; it does not rule out that better-designed probes (e.g. shared-suffix or paired-decorrelated designs) could isolate a genuine representation.

9 Future Work

Natural extensions include: replacing the keyword scorer with a safety-judge model and reporting scorer agreement; extending to additional families and larger scales (14B–70B) to map how far each family’s trend continues; grounding deployment framings in real user corpora rather than controlled rewrites; and designing shared-suffix probes that read activations at an identical continuation to test for genuine internal representations rather than surface separability.

10 Conclusion

Instruction-tuned language models refuse casually-phrased harmful requests less often than identical formally-phrased ones, and—contrary to what an off-the-shelf harm grader would report—this corresponds to a real and substantial increase in actionable harmful output. A keyword refusal metric reports a register gap of 0.27 (up to 0.54 for Qwen-7B); an unvalidated LLM grader reports only 0.03; but validating that grader against 200 human labels reveals it hides most casually-framed harm as “hedging,” and the corrected register harm gap is 0.26 (95% CI [0.19, 0.32]), with casual framing raising the actionable-harm rate from 8.5% to 34.3%. Our central conclusion is methodological and, we believe, broadly relevant as LLM-graded evaluation proliferates: **an unvalidated LLM safety grader can report the absence of a safety problem that in fact exists**, and must be checked against human judgment before its verdicts are trusted. Within the two evaluated model families, the register effect is consistently observed across all tested models; we also find that

prompt-activation probes cannot validly establish internal evaluation-awareness for this distinction. Our study is limited to open models up to 9B and a single grader; extending the human-validated harm measurement to more models, graders, and scales is important future work.

Reproducibility Statement

We release the paired-prompt dataset (all three iterations), the raw model generations for all six models, the refusal labels, and the analysis code at [repository link]. All experiments run on a single consumer GPU in 4-bit precision, making the study reproducible without specialized hardware.

Ethics and Broader Impact

Our prompts describe harmful intents in the abstract solely to measure refusal behavior; we report results at the level of aggregate refusal rates and redact any operationally harmful content from released generations. The register-sensitivity gap we document is a safety-relevant weakness: it indicates that models may comply with casually phrased harmful requests they would refuse under formal testing. We disclose it to improve evaluation practice, not to provide a method for eliciting harmful content; the effect is reported so that evaluators can correct for it.

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